

Introduction to Immunology

Integrated Biomedical and Clinical Sciences OMFS 513

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Learning Objectives

- Compare the general features of immune cells from myeloid lineage.
- Describe the general features of cells from lymphoid lineage.
- Recognize the normal function of immune system and consequences of immune system dysfunction.

Outline

- What is immunology?
- Components of the immune system.
- Functions of the immune system.
- Dysfunction of the immune system.

What is immunology?

Immunology is the study of the immune system.

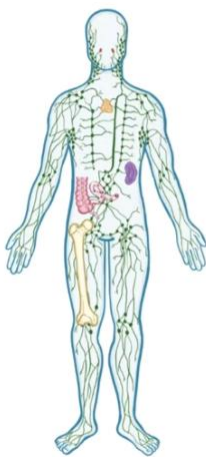
Immunity is derived from the Latin term “immunitas”, meaning “to be exempt from”.

Cells, tissues, organs and molecules responsible for immunity constitute the **immune system**.

Their collective and coordinated response to foreign substances is called an **immune response**.

Lymphatic system

THE LYMPHATIC SYSTEM



LYMPHATIC VESSELS ~ DRAIN INTERSTITIAL FLUID or LYMPH from PERIPHERAL TISSUES into the BLOOD

TISSUES

ORGANS

CONTAIN LYMPHOCYTES & WHITE BLOOD CELLS

PRIMARY LYMPHOID ORGANS

- * THYMUS
- * BONE MARROW



SECONDARY LYMPHOID ORGANS

- * TONSILS
- * LYMPH NODES
- * SPLEEN
- * MUCOSA-ASSOCIATED LYMPHOID TISSUE (MALT)



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Lymphatic System

- Major functions related to immunology:
 - Provide an additional route for the return of extracellular fluid to the circulatory system
 - Help drain fluid that has accumulated due to the inflammatory response.
 - Render surveillance, recognition, and protection against foreign materials through lymphocytes, phagocytes, and antibodies.

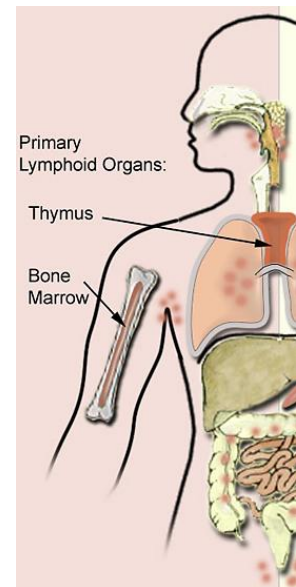
Lymph:

- Plasma-like liquid carried by the lymphatic circulation.
- Formed from interstitial fluid that leaks from blood vessels into the extracellular spaces and migrate into lymphatic capillaries.
- Made up of water (95 %), electrolytes, proteins (antibodies and albumin), fats, white blood cells (mainly lymphocytes) and foreign material that gained access to tissue spaces (antigens).

Primary lymphoid organs

(Generative lymphoid organs: bone marrow & thymus).

- Growth signals for lymphocyte development.
- Have self-antigens to establish central tolerance.
- Myeloid origin: Develop in the bone marrow.
- Lymphoid origin: Develop, mature and are activated in different sites.
 - B cells develop and partially mature in the bone marrow. Then, spleen to complete the maturation and activation.
 - T cells are produced in the bone marrow and mature in the thymus.



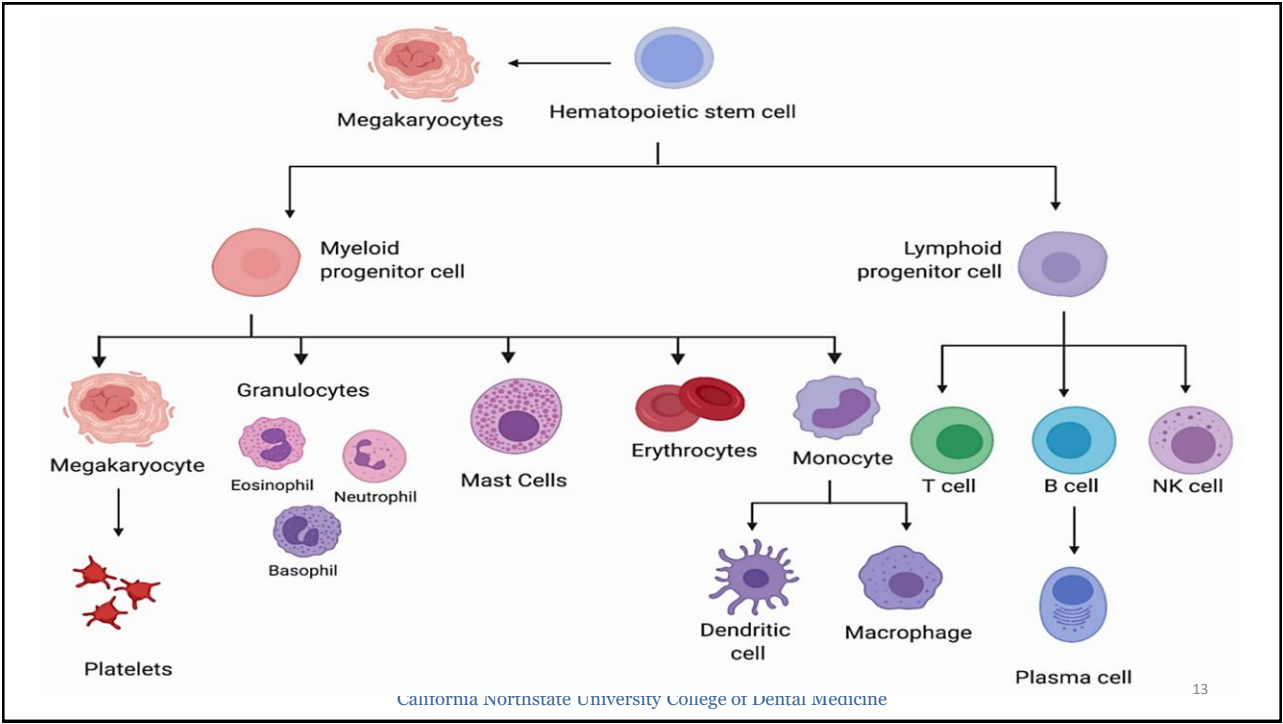
Bone marrow

Red marrow: site for HSCs. Can generate circulating blood cells, including RBCs, WBCs, platelets, lymphoid progenitor cells.

In healthy individuals, the bone marrow contains:

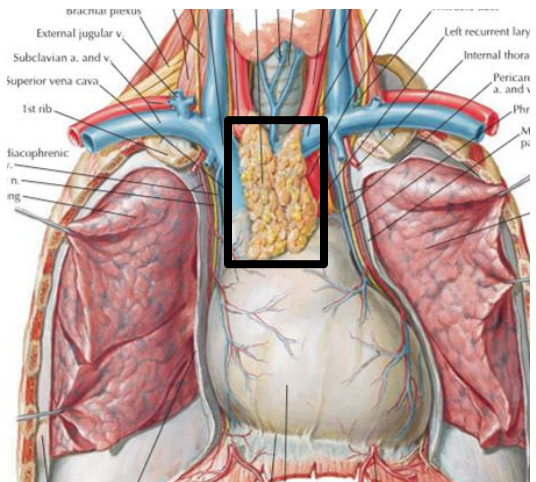
- 70% myeloid lineage.
- 10-15 % lymphoid lineage.
- Non-hematopoietic cells (like adipocytes).

With aging, adipocytes gradually replace $\geq 50\%$ of bone marrow compartment (decreasing hematopoiesis efficiency).

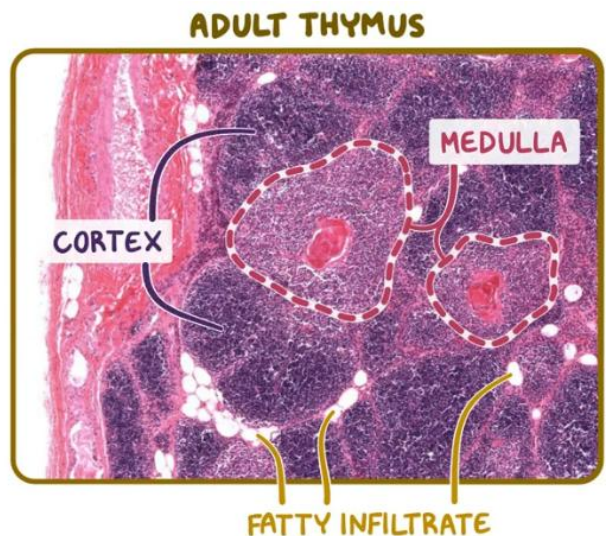


Thymus

- Bi-lobed lymphoid organ encapsulated in a collagenous capsule.
- Located in the anterior part of the superior mediastinum behind the sternum
- Most active during childhood
- After puberty, begin to involute (decrease in size) lymphatic tissue and increase of adipocytes).



- Composed of:
 - basophilic (or darker) cortex regions
 - Inner lighter purple medulla regions.
- Hematopoiesis generates lymphoid precursors that goes to the thymus and become **thymocytes**.
- In the thymus, thymocytes undergo selection, develop into mature, functional but naïve T cells.



Functions:

1- Thymus epithelial cells secrete chemokines to attract lymphoid precursors from blood.
2- Immature T cells generate a unique antigen specific receptors (T-cell receptors or TCRs).

- Immature T-cells begin maturation in the cortex. As thymocytes mature, they migrate toward the medulla.

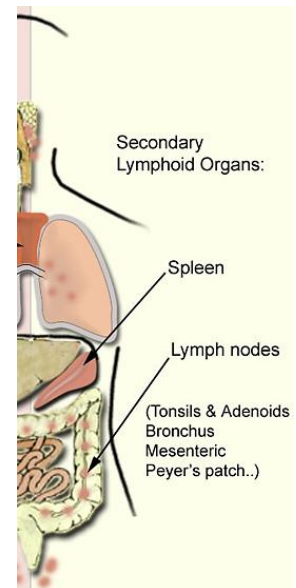
3- To ensure self-tolerance, T-cells are selected based on their reactivity to self antigens “**Positive/negative selection**”. Around 1 % of thymocytes survive this process.

- Positive selection: Occur in the cortex. Thymocytes with active TCR that weakly bind self-antigens survive.
- Negative selection: In medulla. Thymocytes that recognize and bind self-antigens with their TCRs die.

Only mature naive T cells exit the thymus and enter the blood and peripheral lymphoid tissues.

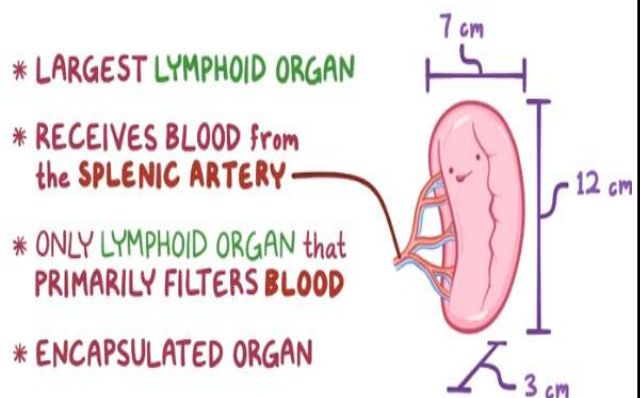
Secondary lymphoid organs

- Spleen, lymph nodes, and components of skin and mucosal lymphatic tissue.
- Responsible for completing maturation and activation of T and B lymphocytes.
- To ensure optimal activation of adaptive immune response, anatomical arrangement of lymph nodes allows the interaction of lymphocytes and antigen presenting cells.



Spleen

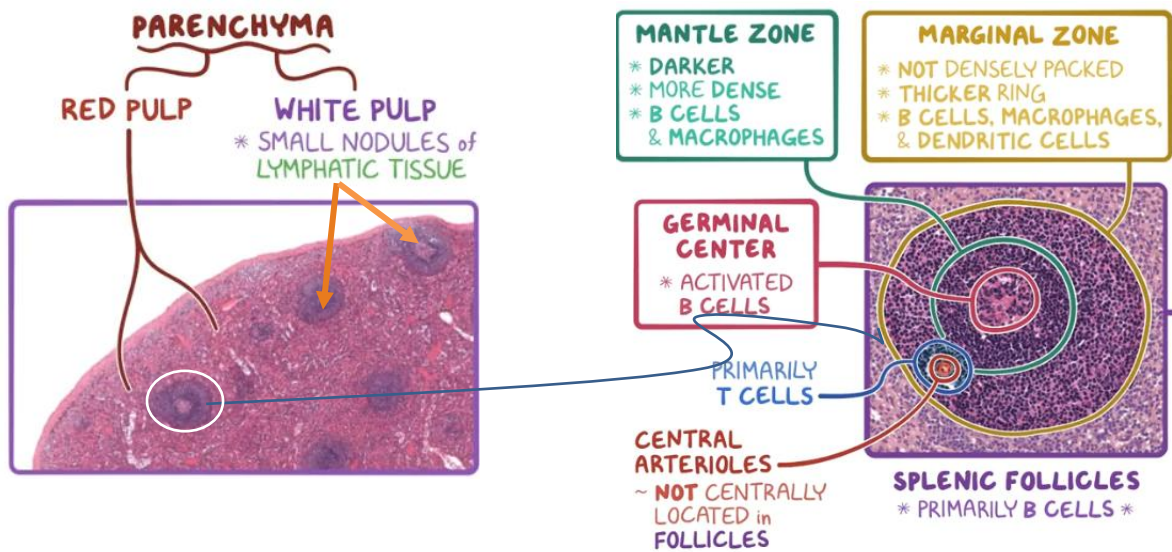
- The spleen is a well-vascularized intraperitoneal organ.
- Functions:
 - remove aged and damaged blood cells and particles (like immune complexes and opsonized microbes) from the circulation.
 - Initiate adaptive immune responses to blood-borne antigens.
 - Iron metabolism.
 - Thymocyte storage.
 - Hematopoiesis (B-cell expansion), B-cell activation.



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Spleen



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Impaired Spleen Function

- Patients without a spleen (asplenia), due to congenital or splenectomy will have defective humoral immunity. They are at higher risk for recurrent infections from encapsulated bacteria (commonly *Streptococcus pneumoniae*, *Haemophilus influenzae* and *Neisseria*) and blood-borne parasites. Patients should receive all vaccines and can receive re-vaccination boosters for certain infections.
- Infection-associated mortality is about two- threefold higher than the general population.
- For certain medical procedures, antibiotic prophylaxis, in addition to vaccination, can further reduce the risk of infection and poor outcomes.
- Dental antibiotic prophylaxis: no consensus regarding the need for antibiotics. The decision should be done in coordination with the physician.

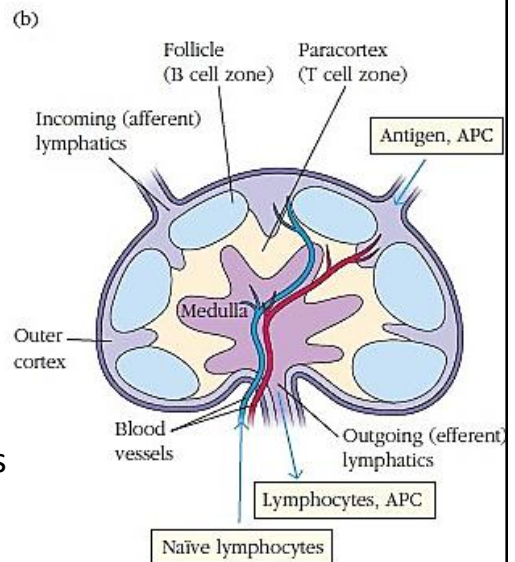
Lymph nodes

Encapsulated, bean-shaped structures connected to both blood and lymphatic vessels.

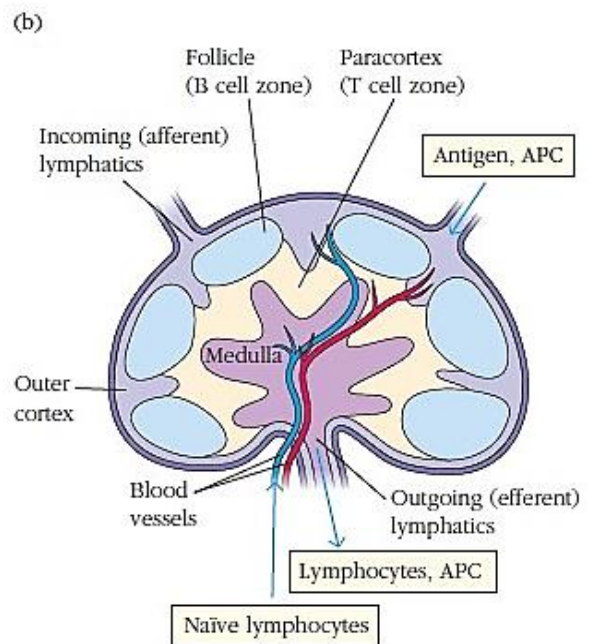
Area where lymphocytes interact with antigen-presenting cells.

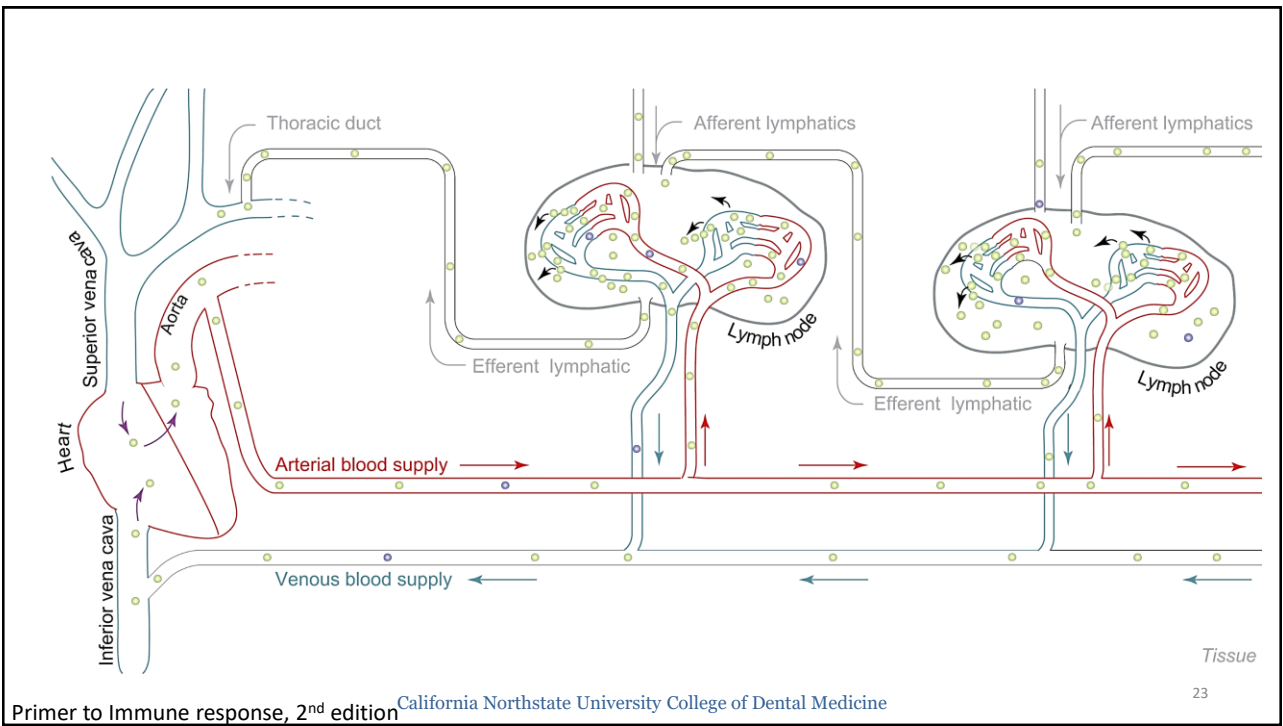
The follicle (cortex), contains B lymphocytes, macrophages, and follicular dendritic cells arranged in follicles.

The paracortex, mostly T lymphocytes and contains dendritic cells that migrated from tissues to the node.



The medulla site where lymphocytes exit the lymph node through efferent lymphatics. Also, some activated plasma cells will reside in the medulla and secrete antibodies to the circulation.





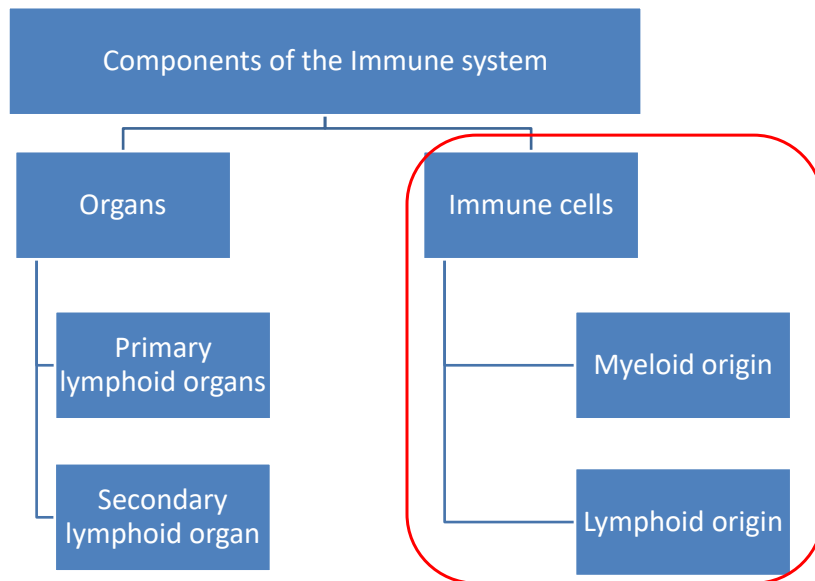
Associated Immune Tissue

SALT, MALT, GALT

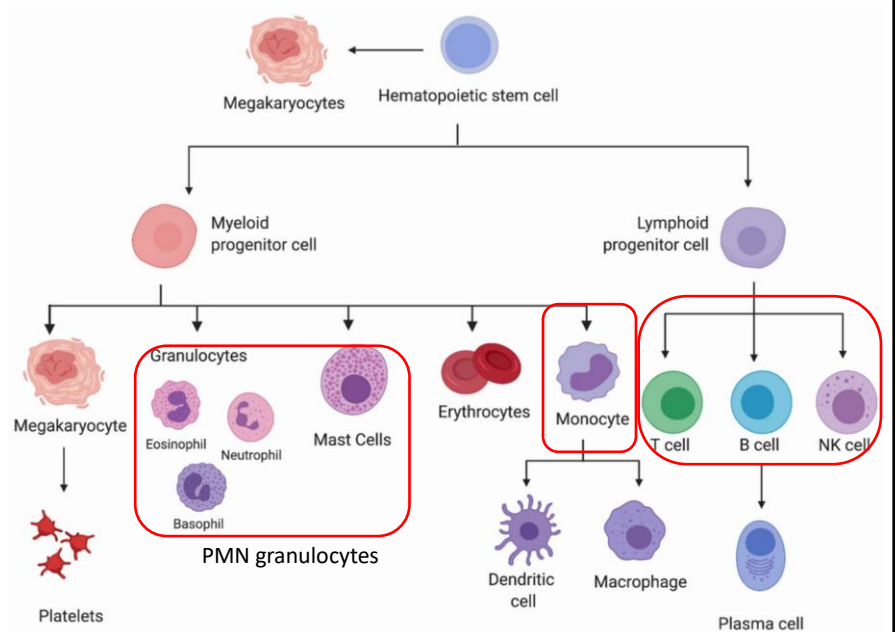
- All are un-capsulated secondary lymphoid tissues, composed of lymphoid follicles.
- MALT: Mucosa-associated lymphoid tissue (e.g tonsils, adenoids), found along the mucosal surface.
 - Inductive sites for the clonal expansion of B-cells and antigen sampling.
 - Effector sites for IgA secretion and immune action.
- GALT: gut-associated lymphoid tissue like Peyer's patches (in the small intestine), appendix or M-cells that sample antigens in the GI and deliver to immune cells.
- Skin-associated lymphoid tissue (SALT) and bronchial-associated lymphoid tissue (BALT).

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Hematopoiesis



Raza Y, Salman H, Luberto C. Sphingolipids in Hematopoiesis: Exploring Their Role in Lineage Commitment. *Cells*. 2021; 10(10):2507. <https://doi.org/10.3390/cells10102507>

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Cells of the Myeloid lineage

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Granulocytes

- Innate immune system.
- Myeloid progenitor lineage can give rise to granulocytes, monocytes and megakaryocytes
- Granulocytes: **neutrophils, basophils, mast cells, and eosinophils.**
- Upon activation, granules release enzymes and inflammatory mediators into the bloodstream.
- Myelopoiesis (development of myeloid cells) starts in the liver in fetus at about 6 weeks of gestation.

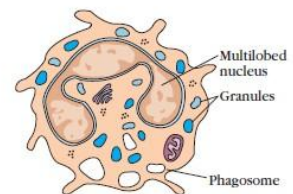
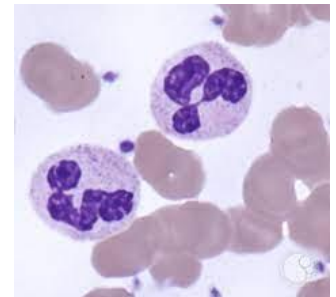
Neutrophils

Granulocyte, multi-lobed nucleus.

Most abundant WBCs. Also termed polymorphonuclear neutrophils (PMNs).

Phagocytes: At site of infection, neutrophils phagocytose (engulf) bacteria and secrete proteins with antimicrobial effect and tissue remodeling potential.

In response to infections, the number of circulating neutrophils increases significantly “leukocytosis”. Dominant first responders to infection, main cellular components of pus, short lifespan of only few days (1-2 days).



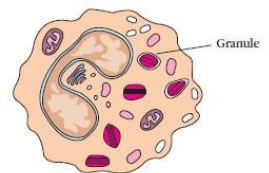
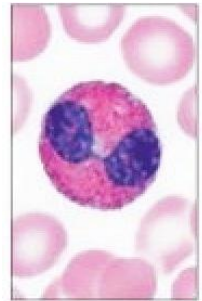
Eosinophils

Granulocyte, large cell size, bi-lobed nucleus.

Eosinophil granules contain basic proteins that bind to acidic dyes, like eosin (Red in stained blood smears).

Motile phagocytic cells

- Instrumental in defense against multicellular parasitic organisms, including worms and fungal infections.
- Cluster around the invading worms and damage their membranes by releasing enzymes from the eosinophilic granules.
- Eosinophils can also contribute to asthma and allergy symptoms.

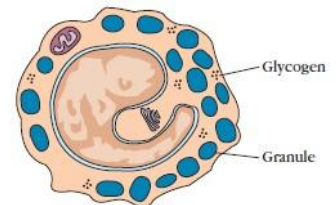
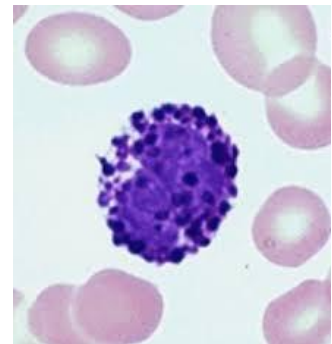


Basophils

Granulocyte, kidney-bean shaped nucleus.

Non-phagocytic blood granulocytes that contain large granules. Granules in **basophils** bind **basic dyes**= **blue** color in stained blood smears.

Stimulated basophils release their granule contents that include **histamine**. Major contributor to allergy symptoms. Increases blood vessel permeability and causes smooth muscle constriction.

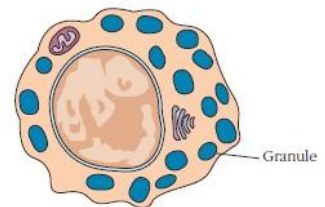
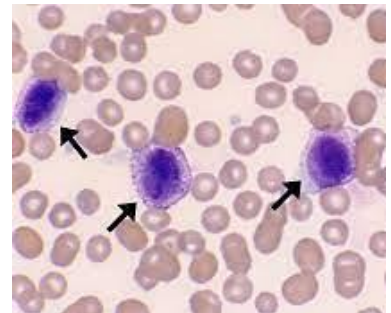


Mast cells

Granulocytes with a large central nucleus.

Non-phagocytic cells with cytoplasmic granules that contains histamine and play an important role in allergic reactions.

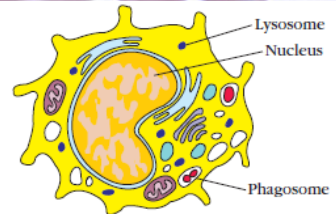
Mature mast cells are adherent to small blood vessels in a wide variety of tissues, including skin, mucosal epithelial tissue of the respiratory, genitourinary, and digestive tracts.



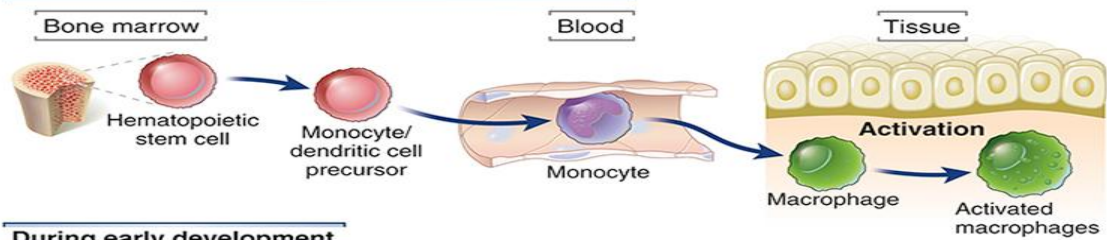
Monocytes

Mononuclear phagocytic cells. Monocytes have kidney bean-shaped nucleus, finely granular cytoplasm and cytoskeletal filaments.

In inflammatory conditions, monocytes can differentiate into macrophages and dendritic cells.

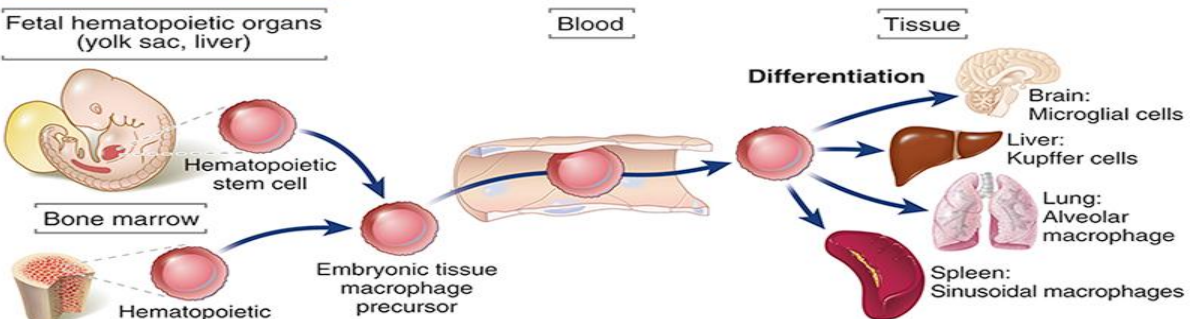


In adult homeostasis and inflammatory reactions



During early development

Fetal hematopoietic organs
(yolk sac, liver)



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Macrophages

Macrophages are mononuclear phagocytes that reside in all organs and tissues.

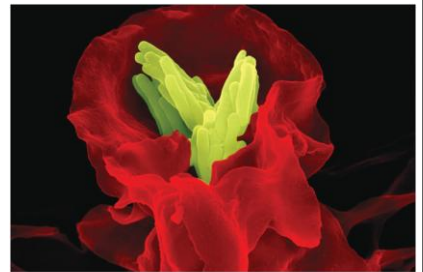
Categorized to:

A) Tissue-resident macrophages:

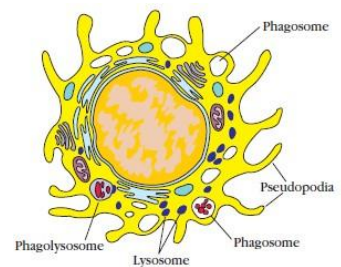
Like, osteoclasts in the bone, microglial cells in the central nervous system, alveolar macrophages in the lung and Kupffer cells in liver.

B) Recruited or monocyte-derived macrophages:

Migrate into tissues during a response to infection.



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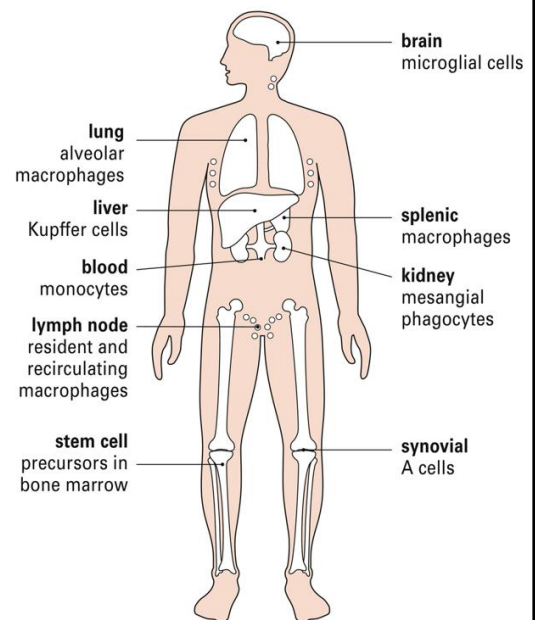
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Macrophages are important in Innate & Adaptive immunity

- Innate immune response, macrophages are important phagocytes. Also, secrete chemokines that attract more leukocytes.
- Bridge to adaptive immune response: macrophages secrete inflammatory cytokines and act as antigen presenting cells to lymphocytes.

Tissue-resident macrophages

- Found throughout the body to intercept pathogens:
 - Heart: cardiac resident macrophages promote angiogenesis, remove ageing and dying cells and participate in cardiac electrical conduction. Play a role in CVS and cardiac physiology.
 - Spleen “red pulp macrophages” to scavenge RBCs and recycle iron.
 - Liver: Kupffer cells
 - Brain: microglia
 - Skin/oral mucosa: Langerhans cells

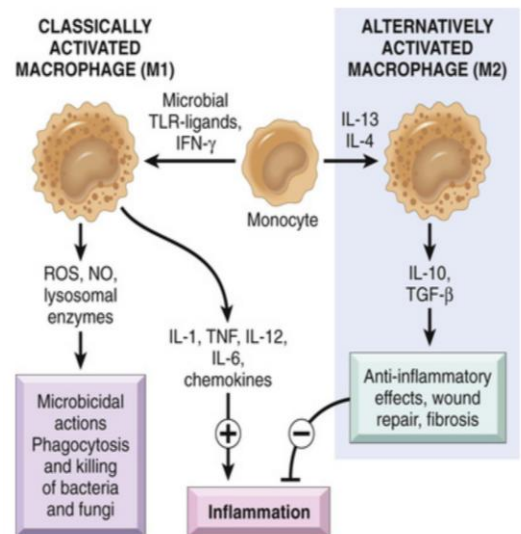


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Recruited or monocyte-derived macrophages: Depending on the immunological encounters:

- 1- Classically activated
“M1 macrophages”: inflammatory macrophages that ingest microbes and activate the adaptive immunity.
- 2- Alternatively activated
“M2 macrophages”: macrophages which have immunosuppressive and tissue remodeling phenotype.

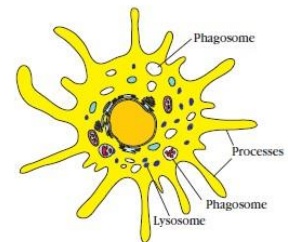


Dendritic cells

Dendritic cells act as a bridge between the innate and adaptive immune responses.

After encountering microbes, dendritic cells become activated and capture the microbial antigen.

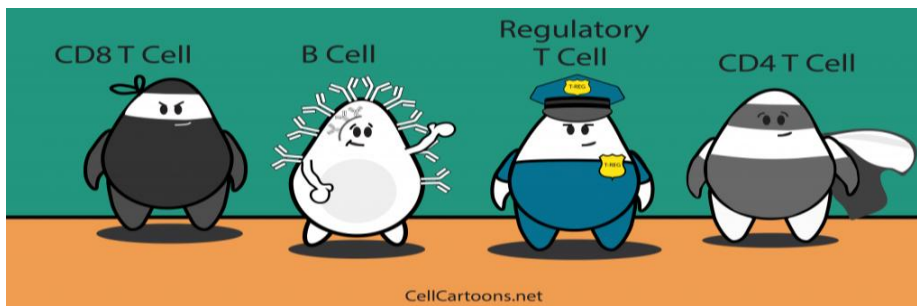
Next, they migrate to lymphoid organs to present the antigens to T-cells which initiates the adaptive immune response.



Myeloid Antigen Presenting Cells

- Myeloid progenitors give rise to a group of phagocytic cells (monocytes, macrophages, and dendritic cells) which function as antigen-presenting cells (APC).
- They can ingest pathogens via phagocytosis, digest pathogen proteins into peptides, then present these peptide antigens on their membrane surfaces (MHC class II).
- Myeloid APCs are cellular bridges between the innate and adaptive immune response.

Lymphoid lineage

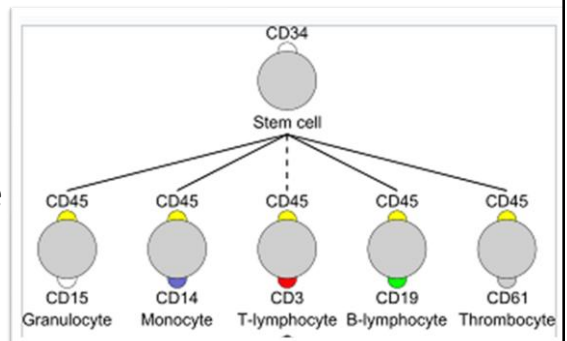


Cluster of Differentiation

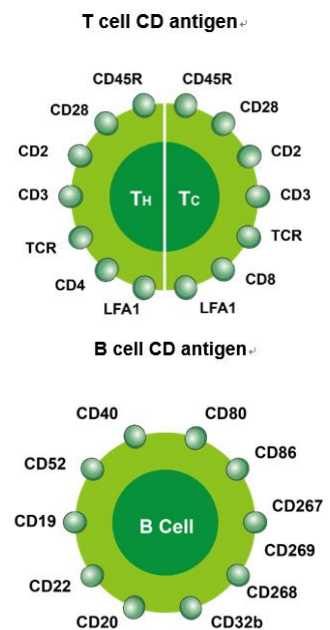
CD are surface antigens that identify and discriminate different cell populations. Each CD is given a unique number (CD1, CD2 till CD371).

These markers can classify different immune cell populations.

They have important functions in detecting the immune response and serves as target for therapeutic antibodies.



- Lymphocytes can be subdivided into two major populations: B lymphocytes (B cells) and T lymphocytes (T cells).
- T and B cells appear similar under a microscope. Therefore, we rely on the expressed CDs to distinguish the lymphocyte subpopulations.
 - ❑ CD4 T cell
 - ❑ CD8 T cell
 - ❑ CD19 and CD20 are specific for B cell life-cycle.
Can be used as markers for B-cell lymphomas

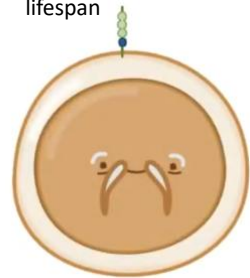


<https://www.creativebiomart.net/Cluster-of-Differentiation.htm>

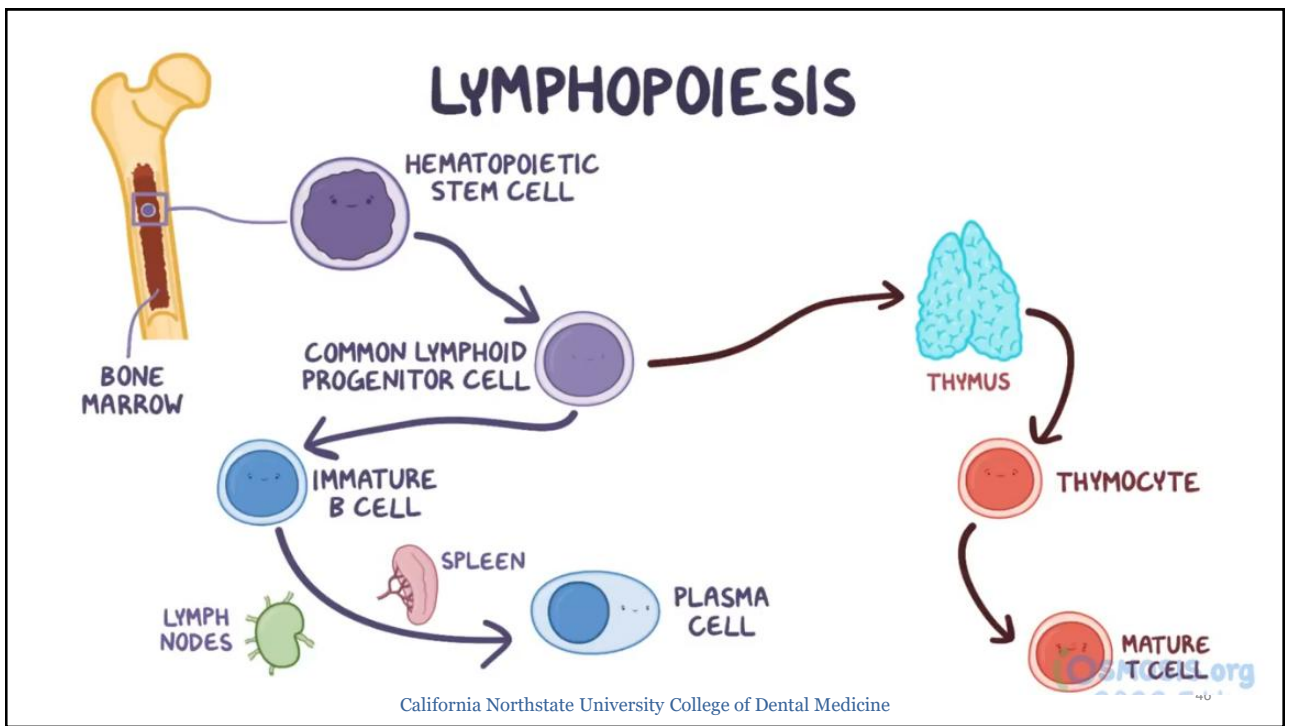
- B cells express an antigen-specific receptor named “B cell receptor”. T cells express the “T cell receptor”.
- Contact with an antigen induce the transformation of naïve lymphocytes to:
 - ☐ Effector cells
 - ☐ Memory cells



Activated Effector cell,
specific function and short
lifespan



Memory cell, long lifespan (>10y).
Immune system archive



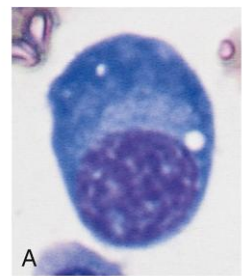
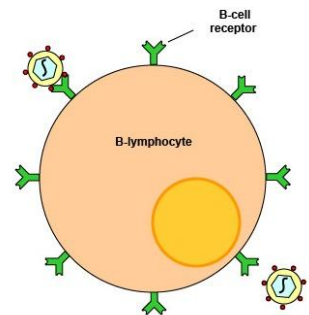
B lymphocytes

B cells are positive for CD45, CD19, CD20 and CD21.

Mature B cells: distinguished from other leukocytes by surface display of B-cell receptor.

The B-cell receptor is a membrane-bound immunoglobulin (antibody) molecule that can bind to antigen. **Capable of antigen presentation**

Each mature B cell expresses approximately $1.5\text{--}3 \times 10^5$ molecules of surface antibodies that have identical binding sites for antigen.



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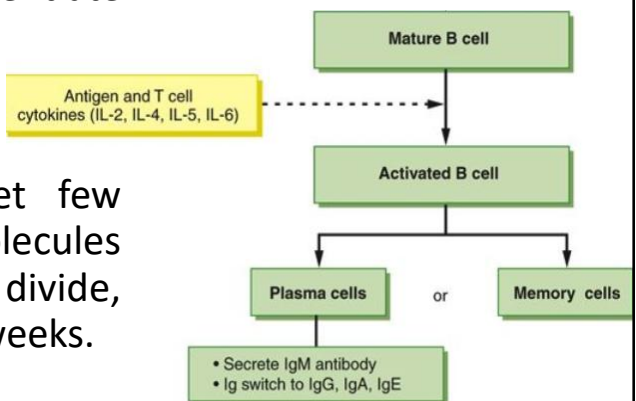
Following activation, B cell differentiate into:

A) Effector cells “plasma cells”

A single plasma cell can secrete few hundred-thousand antibody molecules per second. Plasma cells do not divide, and they live for approximately 6 weeks.

B) Memory B cells

Survive for 10 years and can differentiate to antibody-producing plasma cells.



Louise, Hawley., *Microbiology and Immunology*. Available from: VitalSource Bookshelf, (6th Edition). Wolters Kluwer Health, 2013.

T lymphocytes (T cells)

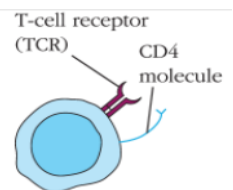
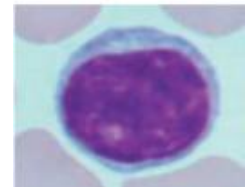
T lymphocytes are divided into three major cell types. All positive for CD3.

- A) **T helper cells**, characterized by the presence of CD4 receptors on surfaces of the cells.
- B) **T cytotoxic cells**, characterized by CD8 receptors on surfaces of the cells.
- C) **Regulatory T cells**, CD4, CD25 receptors on cell surface.

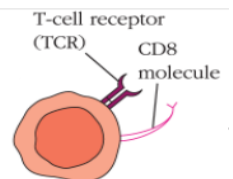
The ratio of CD4 to CD8 T cells is approximately 2:1 in normal physiological conditions.

A change in this ratio is often an indicator of immunodeficiency disease (e.g., HIV), autoimmune diseases, and other pathologies.

(a) Lymphocyte



T helper (T_H) cell



T cytotoxic (T_C) cell

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CD4 T cells

Upon encountering an antigen, naïve CD4 T cells become activated and proliferate.

Differentiate into a variety of effector T cell subsets:

- 1) T helper type 1 (TH1), respond to intercellular pathogens.
- 2) T helper type 2 (TH2), respond to large extracellular pathogens like helminth.
- 3) T helper Type 17 (TH17), respond to extracellular pathogens.
- 4) T follicular helper cells (TFH), stimulate antibody production from B cells.

A subset of the naïve CD4 T cells differentiate into the memory CD4 T cells and those cells will produce more CD4 T cells upon reactivation.

CD8 T cells

Upon activation, naïve CD8 T differentiate into an effector cell name cytotoxic T lymphocyte (CTL).

CTLs eliminate cells that display foreign antigen including virus-infected cells, tumor cells, and cells of a foreign tissue graft.

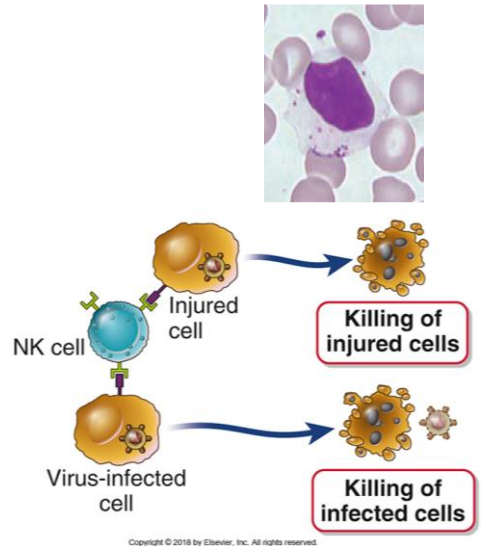
A subset of the naïve CD8 T cells differentiate into the memory CD8 T cells and those cells will produce more CTLs cells upon reactivation.

Natural Killer Cells

Lymphoid cells. Distinguished by the expression of a surface marker known as NK1.1, as well as the presence of cytotoxic granules.

NK cells attack a variety of abnormal cells, including tumor cells and virus-infected cells.

NKT cells are subset of T cells with TCR. Similar function to NK cells + bridge innate & adaptive immune response.



Compartmentalization of the immune cells in the absence of the active immune response

BONE MARROW	BLOOD	TISSUES
HSCs Early hematopoietic progenitors	Monocytes Neutrophils Basophils Erythrocytes	Macrophages Dendritic cells Mast cells NK cells NKT cells Eosinophils
	B lymphocytes T lymphocytes	

Mak, T. W., Saunders, M. E., & Jett, B. D. (2014). *Primer to the immune response*. (second Edition). Elsevier Health Sciences (US).

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Functions of the immune system

Role of the immune system	Implications
Defense against infections	Deficient immunity results in increased susceptibility to infections; exemplified by AIDS Vaccination boosts immune defenses and protects against infections
Defense against tumors	Potential for immunotherapy of cancer
Control of tissue regeneration and scarring	Repair of damaged tissues
The immune system can injure cells and induce pathologic inflammation	Immune responses are the cause of allergic, autoimmune, and other inflammatory diseases
The immune system recognizes and responds to tissue grafts and newly introduced proteins	Immune responses are barriers to transplantation and gene therapy

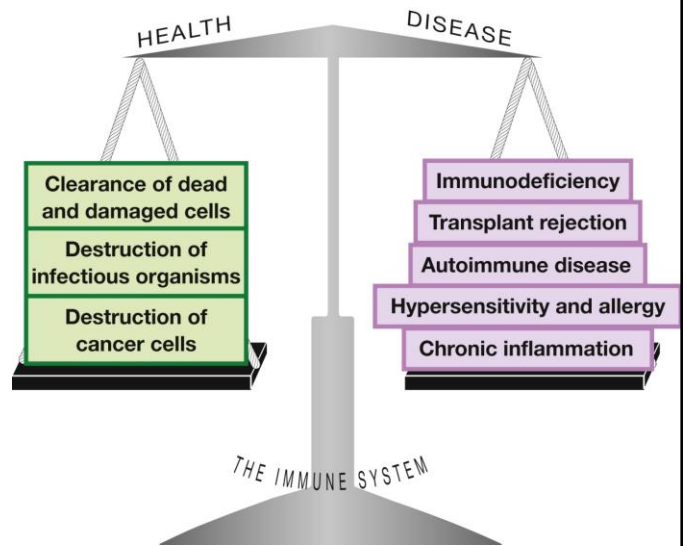
Abbas, Andrew H. Lichtman, Shiv Pillai, A. *Basic Immunology*.
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<https://pubmed.ncbi.nlm.nih.gov/19347848/>
<https://pubmed.ncbi.nlm.nih.gov/26897428/>

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Role of the Immune System in Health and Disease



Mak, T. W., Saunders, M. E., & Jett, B. D. (2014). *Primer to the immune response.* (second Edition). Elsevier Health Sciences (US).

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Dysfunction of the immune system

Hypersensitivity: overreaction of the immune system towards benign antigens. Allergies and asthma are examples of hypersensitivity reactions. These result from an overly reactive immune responses to common environmental antigens (Allergen), such as pollens.

Autoimmune diseases: targeting of self or host proteins by immune cells. In autoimmune diseases, the immune system loses its ability to distinguish between self and non-self or misinterprets a self-component as a source of danger.

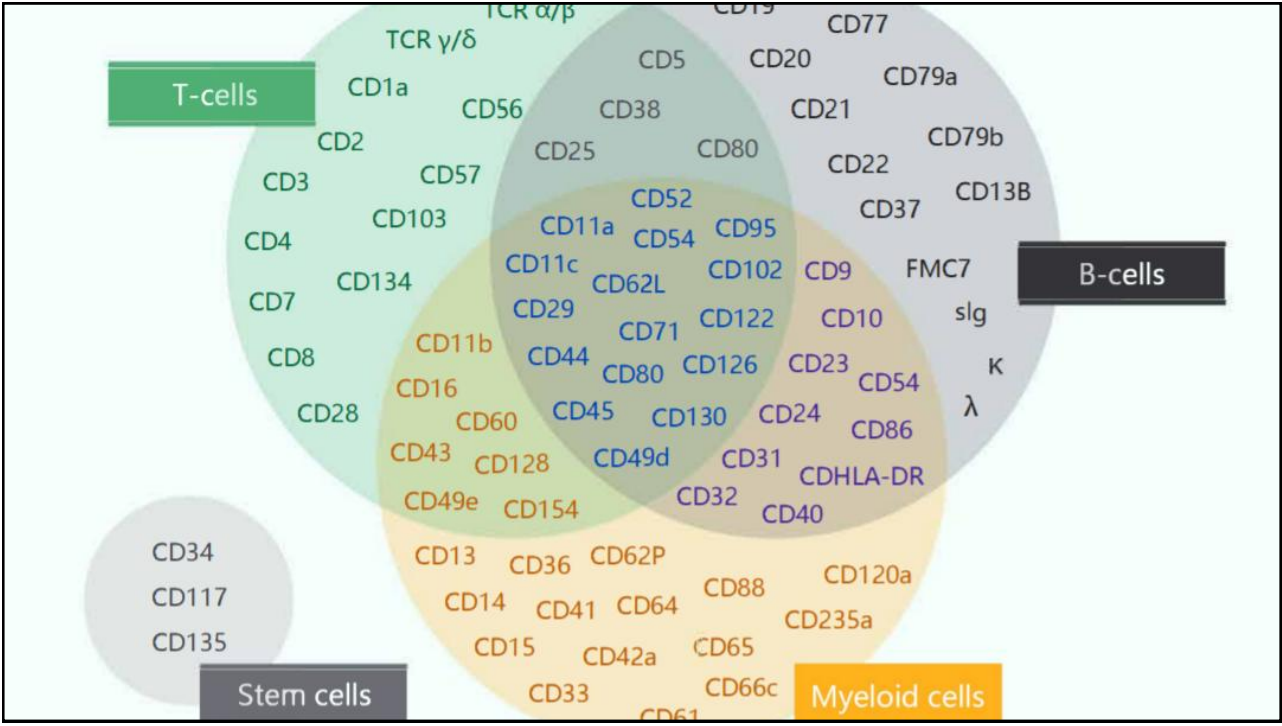
Immunodeficiency: Insufficiency of the immune response to protect against infectious agents. Due to inherited genetic factors “primary immunodeficiencies” like the severe combined immunodeficiency (SCID) . Due to a disruption/damage to the immune system “secondary immunodeficiencies” like in AIDS.

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Role of immune system in periodontal diseases

- Bacteria in the dental plaque provoke immune response (innate and adaptive).
- In response to bacteria, the release of chemokines and cytokines will recruit other immune cells to the area starting the chain reaction.
- Bacterial virulence factors and host immune response lead to the development of inflammatory state, activation of osteoclast, tissue damage and periodontitis.



References

